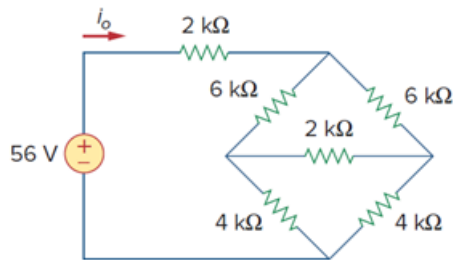


Problem

For the bridge network in Fig. 3.86, find i_o using mesh analysis.

Figure 3.86:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.86 in the text book for the bridge network.

Redraw the circuit by marking mesh currents.

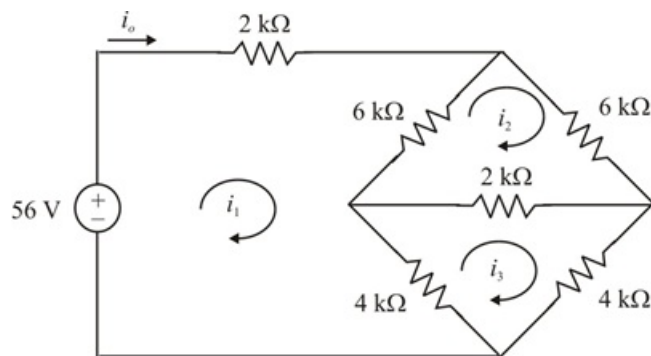


Figure 1

Step 2 of 3

Assume all currents are in mA.

Apply Kirchhoff's voltage law to mesh 1.

$$-56 + (2 \text{ k})i_1 + (6 \text{ k})(i_1 - i_2) + (4 \text{ k})(i_1 - i_3) = 0$$

$$(2 \text{ k} + 6 \text{ k} + 4 \text{ k})i_1 - (6 \text{ k})i_2 - (4 \text{ k})i_3 = 56$$

$$(12 \text{ k})i_1 - (6 \text{ k})i_2 - (4 \text{ k})i_3 = 56$$

$$6i_1 - 3i_2 - 2i_3 = 28$$

Apply Kirchhoff's voltage law to mesh 2.

$$(6 \text{ k})(i_2 - i_1) + (6 \text{ k})i_2 + (2 \text{ k})(i_2 - i_3) = 0$$

$$-(6 \text{ k})i_1 + (6 \text{ k} + 6 \text{ k} + 2 \text{ k})(i_2) - (2 \text{ k})i_3 = 0$$

$$-(6 \text{ k})i_1 + (14 \text{ k})(i_2) - (2 \text{ k})i_3 = 0$$

$$-3i_1 + 7i_2 - i_3 = 0$$

Step 3 of 3

Apply Kirchhoff's voltage law to mesh 3.

$$(4\text{ k})(i_3 - i_1) + (2\text{ k})(i_3 - i_2) + (4\text{ k})i_3 = 0$$

$$-(4\text{ k})i_1 - (2\text{ k})i_2 + (4\text{ k} + 2\text{ k} + 4\text{ k})i_3 = 0$$

$$-(4\text{ k})i_1 - (2\text{ k})i_2 + (10\text{ k})i_3 = 0$$

$$-2i_1 - i_2 + 5i_3 = 0$$

Solve the three mesh equations to get the mesh currents in mA.

$$i_1 = 8\text{ mA}$$

$$i_2 = 4\text{ mA}$$

$$i_3 = 4\text{ mA}$$

The current i_o is equal to i_1 .

Thus, the current i_o in the circuit is 8 mA.